CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

Relaxed actions of monoids: Towards a new cohomology theory

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5 January 2021 (Tuesday), 16:00 (Lisbon time) · Zoom

Abstract

Artin glueings between frames have much in common with semidirect products of groups. If H and N are groups (frames) then the process of constructing semidirect products (Artin glueings) provides a means of finding all objects in which N embeds as a normal (closed) subobject and H embeds as its complement. Indeed these are both orthogonal special cases of a single construction in the category of monoids. In this talk we discuss this construction and the related notion of a 'relaxed action' of monoids. We will explore an analogue of the second cohomology group parameterised by these relaxed actions. We will see that the associated monoid extensions generalise a number of classes of extensions that already exist in the literature. Finally we will discuss how these ideas may be extended to a full cohomology theory.